Agentic Image Restoration: A Structured Review

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Abstract

Multimodal foundation models and autonomous systems are extending image restoration from static forward mappings to closed-loop Agentic Image Restoration (AIR). Conventional CNN, Vision Transformer, and diffusion restorers follow predetermined inference procedures and usually lack explicit degradation diagnosis, runtime tool composition, output verification, or persistent experience. AIR introduces decision processes that observe intermediate states, select and execute restoration actions, assess the result, and may revise subsequent actions or memory. We review representative studies published from 2018 to 2026 and introduce a five-category controller taxonomy comprising Cybernetic Reinforcement Learning (P1), Prompt-Conditioned Restoration (P2), MLLM Reasoning and Tool Use (P3), Memory-Augmented Restoration Agents (P4), and Multi-Agent Restoration Systems (P5). A four-component anatomy of perception, decision, action, and reflection or memory provides an orthogonal descriptive framework. We synthesize evidence across natural photography, 4K super-resolution, autonomous driving, medical reconstruction, remote sensing, and scientific microscopy, followed by a protocol-aware discussion of benchmarks and evaluation.

Keywords: Agentic Image Restoration, AI Agents, Multimodal LLMs, Brain Controller Taxonomy, Reinforcement Learning, Memory-Augmented Restoration, Multi-Agent Restoration Systems, Computational Imaging, Structured Review

1 Introduction

1.1 From Fixed Forward Mapping to Agentic Closed-Loop IR

Image restoration (IR) is a fundamental task in computational vision that seeks to recover an uncorrupted, high-fidelity latent image $\mathbf{x} \in \mathbb{R}^{H\times W\times C}$ from a degraded physical observation $\mathbf{y}$. Historically, this ill-posed inverse problem is modeled as:

$$\mathbf{y} = \mathcal{D}(\mathbf{x}; \boldsymbol{\eta}) + \mathbf{n}, \tag{1}$$

where $\mathcal{D}(\cdot)$ denotes a composite spatial-frequency degradation operator parameterized by physical or environmental factors $\boldsymbol{\eta}$, and $\mathbf{n}$ represents additive measurement noise. Examples of $\boldsymbol{\eta}$ include atmospheric effects, blur kernels, downsampling filters, and sensor attenuation. Practical low-level processing also includes enhancement tasks that improve illumination, contrast, colour, or perceptual quality without assuming a unique degradation model. We use image restoration as a unified term that includes degradation removal, reconstruction, super-resolution, low-light and contrast enhancement, and related quality-improvement tasks. Classical methods rely on mathematical priors and iterative reconstruction, including Total Variation regularisation [1] and block-matching sparse transform filtering [2]. Their analytical assumptions can limit performance on complex and non-stationary real-world corruptions, and some methods are computationally demanding.

The advent of deep learning transformed computational imaging through increasingly capable CNN, Transformer, recurrent, and generative-prior architectures, including SRCNN [3], DnCNN [4], SwinIR [5], Restormer [6], and the convolutional NAFNet [7]. Recent raw-domain, low-light, super-resolution, and event-guided models further broaden this design space [8–15]. Most discriminative restorers substantially improved single-task fidelity and perceptual realism while using a predetermined, often single-pass inference procedure, whereas diffusion-based restorers rely on iterative sampling. Neither design, by itself, necessarily provides an adaptive agentic feedback loop.

Despite these advances, many conventional supervised deep learning models share a structural constraint: the static forward-mapping assumption. Given an input image $\mathbf{y}$, such neural backbones often execute a predetermined feedforward computation:

$$\hat{\mathbf{x}} = f_\theta(\mathbf{y}; \Theta), \tag{2}$$

where network parameters Θ remain static during inference. This open-loop architecture lacks runtime tool composition, output verification, and adaptive error correction. In unconstrained real-world environments with non-stationary, composite degradations, static mapping models can exhibit generalization degradation, over-smoothing, or hallucination artifacts. “All-in-One” unified models, including PromptIR [16] and InstructIR [17], adapt internal features through learned prompts or instructions. Their standard inference procedures operate within a single conditioned model, without external-tool sequence evaluation or post-failure revision.

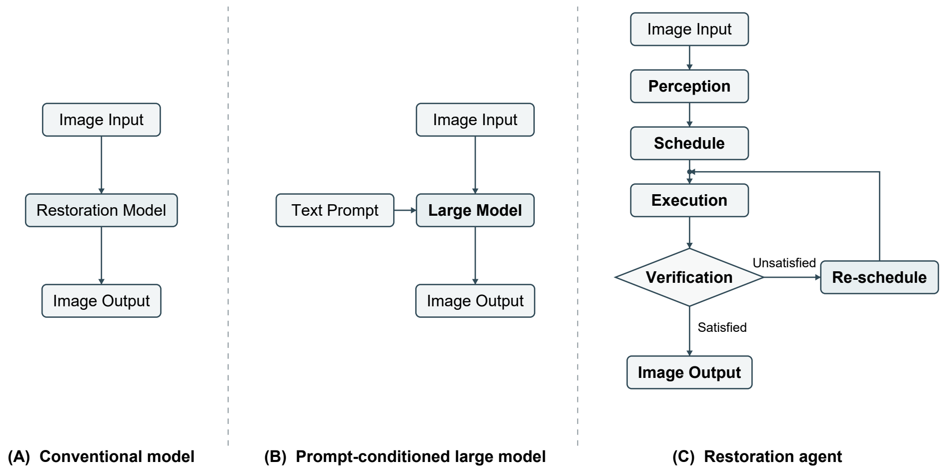

Fig. 1: Conceptual comparison of three image restoration system formulations. (A) Conventional models directly map an input image to a restored output through a fixed restoration model. (B) Prompt-conditioned large models combine the input image with a textual instruction to produce a task-aware restoration result. (C) Restoration agents manage a closed-loop process that perceives image conditions, schedules and executes restoration operations, verifies the resulting quality, and either outputs the image when satisfactory or re-schedules subsequent actions when further improvement is required.

AIR addresses these limitations by reformulating image recovery as an autonomous, goal-directed decision process with feedback, as exemplified by interactive tool-use systems, MLLM restoration agents, and memory-augmented restorers [18–20]. Across the literature, controllers inspect visual states, compose restoration tools, assess candidate outputs, and sometimes retain useful experience. Individual systems implement different subsets of these capabilities, which motivates the controller taxonomy developed below.

Figure 1 summarizes the three inference paradigms introduced above: one-step restoration, all-in-one unified restoration, and agentic restoration. The comparison emphasizes control structure and runtime behavior: an agent manages an adaptive observe–plan–act–verify trajectory with optional retry and memory.

The survey organises Agentic Image Restoration into five Brain Controller Categories: P1 cybernetic reinforcement learning, P2 prompt-conditioned restoration, P3 MLLM reasoning and tool use, P4 memory-augmented restoration, and P5 multi-agent systems. The following sections define these categories and compare their perception, action, reflection, and memory mechanisms.

1.2 Physical Degradation and Agent Decision Models

To establish a solid theoretical foundation, we contrast the classical empirical risk minimization (ERM) framework with the decision-theoretic formulation of Agentic IR.

Conventional deep image restoration formulates training as finding static weights θ^* that minimize an empirical loss over a fixed dataset of degraded-clean pairs $\mathcal{D}_{train} = \{(\mathbf{y}_i, \mathbf{x}_i)\}_{i=1}^N$:

$$\theta^* = \arg\min_\theta \frac{1}{N} \sum_{i=1}^{N} \Big[\mathcal{L}_{pixel}\big(f_\theta(\mathbf{y}_i), \mathbf{x}_i\big) + \lambda \mathcal{L}_{perceptual}\big(f_\theta(\mathbf{y}_i), \mathbf{x}_i\big) \Big]. \tag{3}$$

During test-time inference, the model applies $f_{\theta^*}(\mathbf{y}_{test})$ in a strictly open-loop manner. Equation (3) treats degradations as static mapping targets and typically relies on alignment between training and testing distributions, while providing no explicit mechanism for iterative observation, hypothesis testing, or execution repair.

For comparison across different controller designs, we use a Partially Observable Markov Decision Process (POMDP) as an analytical abstraction. It covers heuristic, prompt-based, and learned controllers by describing the information and actions available during sequential restoration through the seven-tuple

$$\mathcal{M}_{AIR} = \langle \mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \Omega, \gamma \rangle, \tag{4}$$

whose terms are defined below.

- The state space $\mathcal{S}$. The state s_t may include the unobserved latent image $\mathbf{x}^*$, degradation parameters $\boldsymbol{\eta}_t$, spatial error distributions, and system memory. The latent image is a theoretical hidden-state component and is unavailable to the agent at deployment.
- The observation space $\mathcal{O}$. The context o_t is generated through $\Omega(o_t \mid s_t)$. We write $o_t = \langle \hat{\mathbf{x}}_t, \mathbf{e}_t, d_t, \mathbf{q}_t, h_t \rangle$, where $\hat{\mathbf{x}}_t$ is the current image candidate, $\mathbf{e}_t$ is a visual feature, d_t is a language diagnosis when available, $\mathbf{q}_t$ contains image-quality measurements, and h_t is the execution history. Quality measurements can include PSNR, SSIM, LPIPS [21], MANIQA [22], and CLIPIQA [23].
- The action space $\mathcal{A}$. The action a_t is selected from the available tool library.

$$\mathcal{A} = \mathcal{A}_{atomic} \cup \mathcal{A}_{deep\_api} \cup \mathcal{A}_{param\_slider} \cup \mathcal{A}_{code\_gen}, \tag{5}$$

  encompassing mathematical filtering kernels, specialised deep-learning APIs, continuous processing parameters, and scripted workflows. Representative model APIs include SwinIR, Restormer, CodeFormer [24], and NAFNet. The executable set depends on the system under study.
- The transition dynamics $\mathcal{P}$. The distribution $\mathcal{P}(s_{t+1} \mid s_t, a_t)$ describes the state after applying a_t to $\hat{\mathbf{x}}_t$.

- The reward function $\mathcal{R}$. The reward can combine several forms of feedback.

$$\mathcal{R}(s_t, a_t) = \Delta\text{Fidelity}(s_t, s_{t+1}) + \beta\Delta\text{Perceptual}(s_t, s_{t+1}) + \gamma_{task}\Delta\text{Utility} - \mathbf{C}(a_t), \tag{6}$$

  where ΔFidelity measures pixel gains (ΔPSNR), ΔPerceptual measures aesthetic/perceptual improvements (ΔLPIPS, ΔMANIQA), ΔUtility optionally measures downstream task gains (e.g., object detection mAP in driving scenes) when task labels or evaluators are available, and $\mathbf{C}(a_t)$ penalizes execution latency, token expenditure, and tool invocation cost.
- The discount factor γ. The value $\gamma \in [0, 1)$ balances immediate quality gains against later rewards in a multi-step trajectory.

Depending on the system, the agent's Decision Brain may select actions through heuristic prompt routing or a fixed workflow, or it may learn or optimize a policy $\pi^*_\phi(a_t \mid o_t, h_t)$. For learned or optimized policy formulations, the objective is to maximize expected cumulative discounted return across horizon T:

$$\pi^* = \arg\max_{\pi} \mathbb{E}_\pi \left[\sum_{t=0}^{T} \gamma^t \mathcal{R}(s_t, a_t) \right]. \tag{7}$$

1.3 Scope, Selection Criteria, and Key Contributions

This structured review draws on IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, CVF Open Access, OpenReview, arXiv, and Research Square records published from 2018 to 2026. The corpus prioritises primary papers that expose a restoration controller, action interface, feedback mechanism, memory process, or agent-oriented evaluation protocol. PRISMA 2020 provides reporting guidance [25]. The present corpus is organised as a structured technical review with source-traceable selection and verification.

The scope covers natural photography, all-in-one multi-degradation recovery, and 4K super-resolution, represented by 4KAgent [26] and RAR [27]. It also includes autonomous driving and adverse-weather restoration through JarvisIR [28]. Medical imaging is represented by AgentMRI [29], Prompt-Agent PET [30], and Agentic Autoresearch CT [31]. Scientific microscopy is represented by SIMBA [32]. EpiAgent [33] and MAPGR [34] cover cultural-heritage restoration.

This survey makes three contributions.

1. We propose a practical reading framework with four system components and five controller paradigms. The components are Perception and Sensing, Decision Brain, Action and Tools, and Reflection or Memory. The controller paradigms are Cybernetic Reinforcement Learning, Prompt-Conditioned Restoration, MLLM Reasoning and Tool Use, Memory-Augmented Restoration Agents, and Multi-Agent Restoration Systems.
2. We map selected works published between 2018 and 2026 to the taxonomy matrix, recording the reported controller, perception entry point, tool space, and reflection or memory mechanism. The mapping provides a structured snapshot of the reviewed corpus.

3. We discuss possible directions, including Native Physics-World Model Agents, Active Embodied Perception-Restoration, Causal Self-Play & Synthetic Evolution, and Provenance-Certified Proof Agents, as research hypotheses and design opportunities.

2 Unified Taxonomy of Agentic Image Restoration Systems

2.1 The Four-Component Anatomy of Agentic Image Restoration

An agentic image restoration (AIR) system extends a feed-forward mapping into a closed-loop process. Given a degraded image and, when available, auxiliary acquisition information, the system first analyses the degradation and constructs an actionable observation. It then reasons about the restoration objective, plans a sequence of operations, and executes the selected restoration tools. The intermediate or final result is subsequently assessed through quality feedback. The agent may terminate, revise its plan, retry an operation, or roll back to an earlier state, while useful information can be retained for subsequent decisions. Following this processing flow, we organise the anatomy of AIR systems into four components. They are degradation perception and environment sensing, the decision brain, the action and tool space, and the reflection and memory loop.

Module 1. Degradation Perception

Perception is the sensory entry point that translates raw pixel arrays into actionable environment representations o_t. Four complementary mechanisms are used in practice. Implicit feature extraction uses contrastive or prompt-encoder learning to produce a continuous degradation embedding $\mathbf{e}_d \in \mathbb{R}^d$ without explicit semantic labels, as exemplified by DA-CLIP [35]. MLLM semantic diagnosis enables multimodal LLMs to inspect the degraded image and emit a free-text diagnostic report (e.g. "heavy Gaussian noise combined with motion blur"), thereby supporting step-by-step interpretable restoration planning, as pioneered by RestoreAgent [19] and JarvisIR [28]. IQA-guided quality sensing aggregates no-reference metrics (MANIQA [22], MUSIQ [36], CLIP-IQA [23]) and full-reference metrics (PSNR, SSIM, LPIPS [21]) into a quality state vector $\mathbf{q}_t \in \mathbb{R}^k$ that drives learned restore–assess–repeat iterations in RAR [27].

Module 2. Decision Brain

The Decision Brain π_ϕ maps observation o_t and history h_t to action a_t. It is the primary dimension used in the taxonomy of Section 2.2. Implementations range from RL policies with discrete action spaces, as in RL-Restore [37], to MLLM planners, memory-augmented planners such as AgenticIR [20], and multi-agent coordination in MAIR [38] and Hybrid Agents [39].

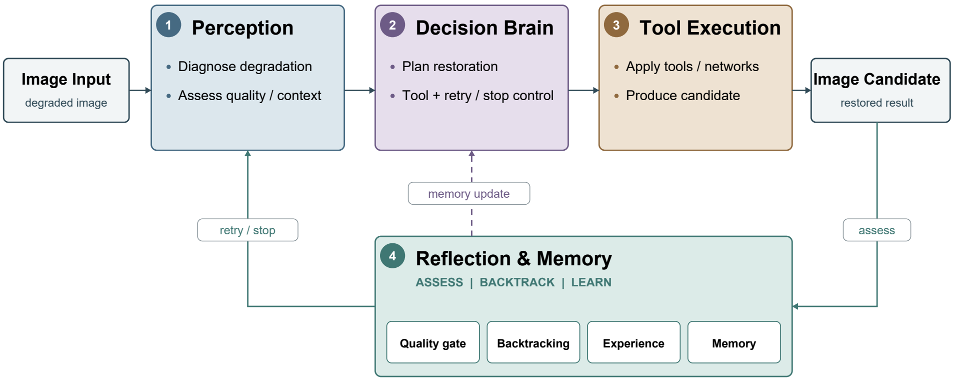

Fig. 2: Four-component anatomy of Agentic Image Restoration (AIR). A degraded input is translated into an actionable observation by the perception module. The decision brain diagnoses, plans, and selects from a heterogeneous tool space. Reflection uses quality feedback to stop, retry, or roll back, while memory records useful experience for future decisions.

Module 3. Action and Tool Space

The tool space $\mathcal{A}$ defines the executable operations available to the Brain. Following the POMDP formulation introduced in Section 1.2, we group its broad capabilities as

$$\mathcal{A} = \mathcal{A}_{\text{atomic}} \cup \mathcal{A}_{\text{deep}} \cup \mathcal{A}_{\text{slider}} \cup \mathcal{A}_{\text{code}}, \tag{8}$$

where the terms denote the following groups.

- $\mathcal{A}_{\text{atomic}}$ contains classical signal-processing kernels such as bilateral filtering, CLAHE, and non-local means denoising.
- $\mathcal{A}_{\text{deep}}$ contains specialised deep-learning model APIs, including SwinIR [5], Restormer [6], NAFNet [7], CodeFormer [24], and StableSR [40].
- $\mathcal{A}_{\text{slider}}$ contains continuous parametric controls such as diffusion sampling steps, guidance scale, or retouching controls available in other image-processing tools.
- $\mathcal{A}_{\text{code}}$ contains scripted image-processing routines that may be available in sandboxed execution environments. Related code-capable-agent benchmarks are reviewed in Section 8.

These groups can overlap. A script may invoke a deep model or expose continuous parameters. The cardinality and heterogeneity of $\mathcal{A}$ can influence the agent's flexibility and the difficulty of policy optimisation.

Module 4. Reflection and Memory Loop

The reflection loop closes the agent's feedback cycle. It evaluates whether the current output $\hat{\mathbf{x}}_t$ meets a quality threshold, decides whether to terminate or to retry with a different tool, and optionally persists knowledge for future use. Four patterns appear in the literature:

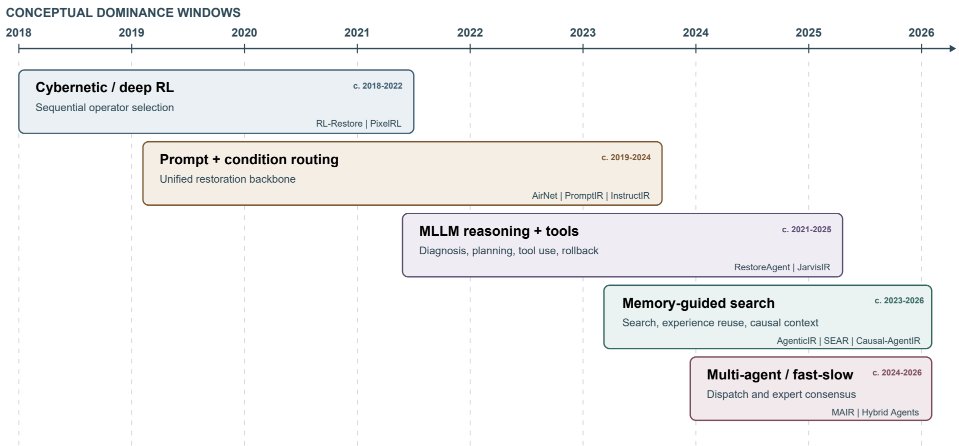

Fig. 3: Representative AIR developments from 2018 to 2026, organised into cybernetic reinforcement-learning policies, prompt-conditioned routing, MLLM tool orchestration, memory-augmented search, and multi-agent collaboration. The temporal overlap reflects concurrent development across controller categories.

- Open-loop output after a single forward pass, as in prompt-routing precursors: PromptIR [16], DA-CLIP [35].
- Quality-guided iteration. RestoreAgent [19] uses execution history to revise model assignments. RAR [27] performs learned assessment and restoration in latent space. These are distinct feedback designs.
- Experience retrieval and reuse. Successful restoration trajectories can be stored in an experience pool and retrieved to inform subsequent decisions, as in AgenticIR [20] and SEAR [41].
- Causal self-evolving memory. A causal graph records operator–degradation interactions. Multi-hop reasoning over the graph guides future decisions and can be updated from restoration experiences. Causal-AgentIR [42] provides a representative example.

2.2 Unified Taxonomy of Agentic Image Restoration Systems

Figure 3 situates representative AIR works from 2018 to 2026. The literature is organised into five primary Brain Controller Categories, P1–P5. These categories follow controller design, so their development periods may overlap. The categories are mutually intelligible at the level of the dominant controller, although hybrid systems may expose attributes of more than one category.

Category P1. Cybernetic Reinforcement Learning

The primary decision maker is a policy $\pi_\phi(a_t|s_t)$ optimised through reinforcement learning. Depending on the source, actions select restoration tools, tune reconstruction parameters, control pixels, route regions, or choose acquisition steps. Rewards may use full-reference quality, reconstruction fidelity, computational cost, or MLLM and no-reference quality signals. Policy optimisation defines this category, while evaluator modality may vary.

Category P2. Prompt-Conditioned Restoration

A learned degradation representation, visual prompt, or textual instruction conditions a shared restoration backbone. The condition changes internal feature processing so that one network can handle multiple degradation types.

Typical prompt-routing engines use one-pass conditioning. Their standard implementations omit explicit output-quality evaluation, retry logic, and cross-image memory, leaving multi-step or composite-degradation errors without an internal correction loop.

Category P3. MLLM Reasoning and Tool Use

A centralised Multimodal LLM (GPT-4V, LLaVA, InternVL, or a fine-tuned variant) commonly serves as the Decision Brain. It can read the degraded image, produce a natural-language diagnosis, decompose the task into a tool sequence, and invoke specialised restoration APIs. Depending on the system design, intermediate results may also be evaluated and used for reflection, replanning, or rollback.

Category P4. Memory-Augmented Restoration Agents

Long-term structured memory supplements the MLLM brain and can support cross-image and cross-task knowledge accumulation. Systems may retrieve successful trajectories from experience pools or represent operator–outcome interactions in structured memory, such as causal graphs.

Category P5. Multi-Agent Restoration Systems

Intelligence is distributed across multiple specialised agents. A dispatcher coordinates specialist agents, while some systems combine a lightweight FastAgent for clear requests, an MLLM-based SlowAgent for ambiguous requests, and a FeedbackAgent for quality assessment. Heterogeneous expert committees and multimodal fusion councils are representative organisational forms.

The five controller categories define the taxonomy. The remaining descriptors are comparison attributes. Generated-code execution can be present in a P3, P4, or P5 system without changing its controller class. A system can mix controller features (e.g. SEAR combines P3 MLLM reasoning with P4 episodic memory), but it is assigned to the category corresponding to its primary decision mechanism. Benchmark suites and evaluation protocols serve as evaluation resources and are reviewed in Section 8.

Generated-code execution is a representative orthogonal capability. A P3–P5 controller may synthesise, compile, and debug Python/PyTorch algorithms in a sandboxed environment as an alternative to a fixed API registry. This capability broadens the Action / Tool Space while retaining the controller category defined above.

2.3 Literature Taxonomy Matrix

Tables 1–3 present a cross-referencing matrix for representative restoration works spanning all five controller paradigms and multiple application domains from 2018 to 2026. Evaluation suites are discussed separately because their role is to assess agent behaviour. Each row encodes the Brain Paradigm together with the Perception mechanism, Action/Tool space type, Reflection/Memory depth, venue, and year of publication.

Table 1: Architectural taxonomy of representative AIR works from 2018 to 2026. Part I covers cybernetic reinforcement learning and prompt-conditioned restoration. P1–P5 denotes the dominant controller paradigm.

Work	**P**	**Venue/ year**	**Perception**	**Action**	**Feedback/ memory**	**Domain**
RL-Restore [37]	P1	CVPR 2018	CNN state features	Discrete operator pool (12)	PSNR reward and stop action	Natural
Distort-Recover [43]	P1	CVPR 2018	Global image state	Discrete global colour actions	Unpaired distort-and-recover reward	Photo retouching
PixelRL [44]	P1	AAAI 2019	Pixel-wise spatial context	Pixel-level discrete actions	Multi-step pixel reward, episode local	Natural
Path-Restore [45]	P1	TPAMI 2022	Path-complexity router	Sub-network routing paths	Efficiency gain signal (Episode)	Natural
CT-DRL [46]	P1	TMI 2018	Iteration-state features	Recon. parameters	Scalar quality reward, episode local	Medical CT
MRI-RL [47]	P1	Neurocomp. 2024	Dual k-space states	Sampling trajectory actions	SSIM reward, episode local	Medical MRI
AirNet [48]	P2	CVPR 2022	Contrastive degradation embedding	Single-backbone feature modulation	Open loop	Natural
PromptIR [16]	P2	NeurIPS 2023	Input-conditioned learned prompts	Prompt-modulated Transformer	Open loop	Natural
InstructIR [17]	P2	ECCV 2024	NL instruction parsing	Instruction-conditioned network	Open loop	Natural
DA-CLIP [35]	P2	ICLR 2024	CLIP content and degradation embeddings	Content and degradation control	Open loop	Natural
AutoDIR [49]	P2/B	ECCV 2024	BIQA degradation ID or text	Structure-corrected latent diffusion	Iterative BIQA routing	Natural
OneRestore [50]	P2	ECCV 2024	Scene and degradation description	Composite-degradation restorer	Open loop	Natural

Table 2: Architectural taxonomy of representative AIR works from 2018 to 2026. Part II covers single-MLLM controllers and learned boundary cases.

Work	P	Venue/ year	Perception	Action	Feedback/ memory	Domain
Clarity ChatGPT [18]	P3	Preprint 2023	ChatGPT + CLIP degrad. detection	Classical + DL tool set	Iterative user feedback (Episode)	Natural
RestoreAgent [19]	P3	NeurIPS 2024	MLLM degradation report	Scored DL API registry	History-based replanning and rollback	Natural
JarvisIR [28]	P3	CVPR 2025	Driving VLM diagnosis	Weather operator pipeline	IQA reward and perception evaluation	Driving
4KAgent [26]	P3	NeurIPS 2025	VLM and IQA perception	Quality-driven mixture of experts	Recursive execution and reflection	Super-res.
RAR [27]	P2/B	CVPR 2026	Learned latent assessor	End-to-end latent restoration	Restore–assess–repeat	Natural
AgentMRI [29]	P3	JIIM 2026	Medical VLM classification	Three CycleGAN correction paths	Consensus gate and stop decision	Medical MRI
EpiAgent [33]	P3	CVPR 2026	Observe–conceive loop	Inscription restoration tools	Execute–reevaluate refinement	Cultural relics
SIMBA [32]	P3	Preprint 2026	Natural-language scientific description	JSON tool-call sequences	Execution verification, episode local	Microscopy
FAPE-IR [51]	P3	CVPR 2026	Frequency-aware planner	High/low-frequency LoRA experts	Planner-conditioned execution	Natural
Restore-R1 [52]	P1/B	CVPR Findings 2026	Frozen visual encoder	RL-learned tool selector	DeQA-Score increment	Natural

Table 3: Architectural taxonomy of representative AIR works from 2018 to 2026. Part III covers memory-augmented and multi-agent controllers.

Work	**P**	**Venue/ year**	**Perception**	**Action**	**Feedback/ memory**	**Domain**
AgenticIR [20]	P4	ICLR 2025	Multimodal diagnosis	Specialised tool registry	Depth-first rollback and experience	Natural
SEAR [41]	P4	Preprint 2026	Episodic fingerprint	Intuitive executor and deliberate planner	P-MCTS with episodic memory	Natural
Causal-AgentIR [42]	P4	Preprint 2026	Quality, degradation, history	Causal memory graph	Persistent graph updating	Natural
MAIR [38]	P5	IJCV 2026	Scheduler	Seven specialist agents	Sequential scheduling and candidate comparison	Natural
Hybrid Agents [39]	P5	CVPR 2026	Fast, Slow, and Feedback agents	Request-dependent restoration	Feedback-based selection	Natural
MAPGR [34]	P5	npj Heritage Sci. 2026	Visual reasoning, prompt, and execution roles	Residual diffusion	Evidence propagation and abstention	Cultural relics

3 Cybernetic Reinforcement Learning

3.1 Foundational MDP Formulation of Cybernetic RL Agents

Early reinforcement-learning (RL) restoration systems introduced explicit states, actions, rewards, and termination decisions. They formulate image recovery as a Markov Decision Process (MDP) governed by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma)$ instead of a static mapping $f_\theta(\mathbf{y}) = \hat{\mathbf{x}}$. A policy π_ϕ observes the degraded image state $\mathbf{s}_t \in \mathcal{S}$, selects an action $a_t \in \mathcal{A}$, and receives the updated image $\mathbf{x}_{t+1} = \mathcal{T}(\mathbf{x}_t, a_t)$. Systems with a learned stop action terminate when the policy predicts that further processing is unnecessary.

The main design variables are the structure and spatial granularity of the action space $\mathcal{A}$. In a discrete MDP, the agent selects an index from a predefined catalogue of lightweight operators. In a continuous MDP, the action specifies numerical parameters such as colour-curve control points, regularisation weights $\lambda(x)$ in iterative CT reconstruction, or smoothing coefficients (σ_s, σ_r) in bilateral filtering. Spatial scope also varies. Some policies apply global actions, while PixelRL operates at pixel level and Path-Restore routes image patches through branches of different depth.

Policy learning is driven by a stepwise quality reward $r_t = \mathcal{R}(\mathbf{s}_t, a_t, \mathbf{s}_{t+1})$, which can quantify incremental gains in restoration metrics or physical fidelity. Some systems also include computational cost or acquisition constraints. This closed-loop evaluation allows the policy to adapt its trajectory to the observed state and establishes the control formulation used by the systems discussed in Section 3.2.

3.2 Representative Systems and Methodological Evolution

We compare representative RL systems through four control designs. These are global discrete tool selection, spatially adaptive routing, continuous parameter tuning, and dual-state active acquisition.

RL-Restore [37] treats restoration as a discrete multi-step decision problem. A Q-learning agent observes global image features, selects from a fixed toolbox of lightweight CNNs for blur, noise, and compression, and may emit a stop action. The reward is based on stepwise PSNR gain. Its action selection operates at image level and therefore applies uniform control within each image.

PixelRL [44] assigns an agent to each pixel in a fully convolutional A3C architecture. Reward Map Convolution incorporates neighbouring future states into the reward calculation to coordinate local decisions. Path-Restore [45] uses a Pathfinder module trained with a difficulty-regulated reward. It routes easier patches through shallow paths and harder patches through deeper residual paths, trading restoration quality against computation.

Discrete action spaces can still support fine-grained sequential adjustment. Distort-and-Recover [43] uses discrete global colour-editing actions and learns from synthetically distorted high-quality images, avoiding paired input and retouched-target training data. Medical inverse problems instead expose reconstruction or filtering parameters as the action interface. Shen *et al.* [46] use a parameter-tuning policy for iterative CT reconstruction. Patwari *et al.* [53] apply reinforcement learning to joint bilateral filtering parameters under an image-quality reward. These systems retain an explicit processing interface while adapting parameters to the observed reconstruction state.

Online reconstruction can make active sampling expensive when a deep reconstructor is evaluated at every decision. DUAL [47] separates a hidden DNN state used to compute training rewards from a visible state used by the online phase-selection policy. The source reports 9 ms decision time, compared with 1363 ms and 1333 ms for two reference methods, corresponding to an approximately 150-fold speed-up in that experimental setting.

PaAgent [54] combines an RL tool-selection policy with subjective and objective feedback. Its name refers to a portrait-aware bank of restoration tools. The reported action space contains 14 restoration tools and covers 14 degradation scenarios. Group Relative Policy Optimisation trains the policy using MLLM semantic assessment together with no-reference image-quality signals. The MLLM therefore contributes reward information while the learned policy remains the action selector.

4 Prompt-Modulated and Condition-Routing Systems

P2 systems broaden restoration coverage by conditioning a shared network on a learned degradation representation, a prompt, or a natural-language instruction. They expose an explicit control signal that adapts the restoration network. In most cases, this signal modulates the backbone within one inference procedure, without a verified sequence of external tool calls.

4.1 Implicit and Learned Degradation Prompts

AirNet [48] learns a degradation representation contrastively and uses it to support all-in-one restoration. PromptIR [16] introduces learned prompts into a Transformer restoration backbone, allowing degradation-dependent feature modulation. MPerceiver [55] extends multi-task restoration with a multi-modal prompt-perception design. These models may adapt computation inside the network through model-internal routing and conditioning.

4.2 Language and Vision–Language Conditioning

InstructIR [17] conditions restoration on human instructions, making the requested operation explicit. DA-CLIP [35] learns degradation-aware representations in the representation space of CLIP [56] for restoration. AutoDIR [49] uses BIQA to identify a predominant degradation and can apply its structural-corrected latent-diffusion restorer iteratively, while OneRestore [50] represents scenes and degradations for composite restoration. Their conditioning mechanisms differ and are therefore described separately.

Language-driven adverse-weather restoration (LDR) [57] uses a pre-trained vision–language model to generate description-based priors for degradation occurrence, type, and severity, then sparsely selects restoration experts in a mixture-of-experts network.

SPIRE [58] uses content prompts and language-specified restoration strength to control a generative restoration model. UniRes [59] addresses complex degradations by combining specialised models during diffusion sampling and exposing a fidelity–quality trade-off. These systems use model-internal routing or conditioning mechanisms.

SUPIR [60] and DreamClear [61] use generative priors for real-world restoration and can use semantic conditions. Their perceptual-quality objectives differ from full-reference PSNR-oriented restoration. Numerical comparison therefore requires a common dataset and metric implementation.

4.3 P2–P3 Boundary

The classification criterion is the locus of control. When a prompt is encoded once and the restoration backbone produces an output without explicit observe–act–verify decisions, the system is P2. AutoDIR is treated as P2/B because its closed-set BIQA module can repeatedly select a dominant degradation within a fixed routing space. If an MLLM diagnoses the image, invokes external tools, examines results, and can revise the plan, it is P3 or above. RAR [27] is an iterative learned boundary case based on internal latent control.

P2 establishes that a single backbone can be controlled by an interpretable or learned condition. Runtime tool orchestration, auditable retry, and persistent experience define the transition to higher-level agentic controllers.

Table 4: Mechanism-level comparison of representative prompt-conditioned systems. Pooled PSNR is omitted because the sources use different tasks and protocols.

Work	Venue/ year	Conditioning signal	Restoration interface	Runtime loop
AirNet [48]	CVPR 2022	Contrastive degradation representation	Unified restoration network	Open
PromptIR [16]	NeurIPS 2023	Learned degradation prompts	Prompt-modulated Transformer	Open
InstructIR [17]	ECCV 2024	Natural-language instruction	Instruction-conditioned network	Open
DA-CLIP [35]	ICLR 2024	Degradation-aware CLIP representation	Conditioned restoration network	Open
MPerceiver [55]	CVPR 2024	Multi-modal prompt perception	Multi-task restoration network	Model-internal
AutoDIR [49]	ECCV 2024	BIQA-inferred degradation or user text	Structural-corrected latent diffusion	Iterative BIQA routing
OneRestore [50]	ECCV 2024	Scene/degradation representation	Composite-degradation restorer	Open
LDR [57]	CVPR 2024	Vision–language degradation description	Sparse MoE restoration network	Model-internal
SPIRE [58]	ECCV 2024	Content and restoration-strength text prompts	Controlled generative restorer	Open
UniRes [59]	ICCV 2025	Complex-degradation formulation	Specialists combined during diffusion sampling	Model-internal
SUPIR [60]	CVPR 2024	Semantic/perceptual condition	Generative restoration model	Open
DreamClear [61]	NeurIPS 2024	Real-world degradation/semantic condition	Generative restoration model	Open

5 Single-MLLM Reasoning and Tool-Orchestrating Agents

Multimodal large language models (MLLMs) enable restoration systems to express a degradation diagnosis, select specialised tools, and interpret intermediate outputs. We use P3 for systems in which one language or vision–language controller is the principal runtime decision maker. Prompt-conditioned single-pass networks remain P2. Learned iterative models such as RAR [27] form a boundary group because their assessment and restoration occur end-to-end in latent space.

5.1 Explicit Diagnosis and Task-Coupled Perception

RestoreAgent [19] made degradation identification and restoration-model assignment explicit, selecting an ordered sequence of degradation types and corresponding models under a scoring function. JarvisIR [28] extends this idea to autonomous driving, where restoration is evaluated through downstream perception utility. On its CleanBench-Real evaluation, the paper reports a 50% average improvement across the perception metrics relative to using the degraded input.

4KAgent [26] combines a profiling module, a VLM-and-IQA perception agent, and a restoration agent that uses quality-driven mixture-of-experts selection in a recursive execution–reflection process. Domain-specific systems adapt the observation interface to acquisition physics or scientific semantics: AgentMRI [29] targets accelerated MRI, EpiAgent [33] cultural-inscription restoration, and SIMBA [32] natural-language control of single-molecule imaging workflows. The Research Square preprint describing FMIRAgent [62] reports an MLLM-based microscopy assistant that identifies degradation, selects or adapts restoration algorithms, explains its choices, and refines decisions using user or simulated feedback. RIR-Agent [63] applies MLLM perception, LLM planning, and a predefined specialist toolbox for composite remote-sensing degradation. Its executor supports adaptation and rollback, and a separate manual mode provides user-customised workflows with real-time MLLM feedback. Their inputs, feedback signals, and success criteria differ, so quantitative results are reported within the setting defined by each source.

5.2 Planning and Tool Selection

RestoreAgent expresses tool selection as

$$\sigma^* = \arg \max_{\sigma \in \mathcal{S}(\mathcal{D},\mathcal{M})} S(I, \sigma), \tag{9}$$

where $\mathcal{S}(\mathcal{D}, \mathcal{M})$ is the set of valid degradation–model sequences and S is the chosen scoring function [19]. Subsequent systems modify the planning space or the coupling between planning and execution.

- Q-Agent [64] combines robust multimodal degradation perception with quality-driven greedy restoration.
- DiTTo [65] models degradation order and uses a simulator to reduce the cost of constructing order-aware action trajectories.
- OPERA [66] jointly optimises the restoration plan and the specialised executors, addressing the mismatch between an independently trained planner and tool pool.
- FAPE-IR [51] combines a frozen MLLM planner with a frequency-aware LoRA mixture-of-experts diffusion executor.
- Restore-R1 [52] is an adjacent RL boundary case. Its lightweight actor–critic policy is trained with DeQA-Score improvement using PPO-style optimisation, with GRPO evaluated as an alternative, as detailed in Section 3.

Creative enhancement has related but distinct objectives. PhotoArtAgent [67] uses language-model artist agents for iterative photo retouching. PhotoAgent [68] plans long-horizon aesthetic edits by tree search with memory and visual feedback and introduces UGC-Edit. RetouchIQ [69] controls Lightroom for instruction-based retouching and trains MLLM agents with a generalist reward. These systems optimise instruction adherence and aesthetic preference, requiring evaluation separate from full-reference restoration fidelity.

5.3 Verification, Iteration, and Boundary Cases

P3 feedback ranges from execution-history-based retry in RestoreAgent to downstream perception feedback in JarvisIR and task-specific review in EpiAgent. The methods use different assessors, stopping rules, and revision procedures. An iterative architecture is also distinct from an MLLM tool controller. RAR [27] is a fully trainable restore–assess–repeat framework operating in latent space and is included as a learned boundary case.

The main synthesis is architectural. Explicit diagnosis can make tool choice interpretable. Task-coupled feedback can align restoration with downstream utility. Joint planner–executor training can improve coordination. The literature uses different tools, test mixtures, stopping rules, and quality metrics. Quantitative conclusions are therefore reported within each source-defined evaluation setting.

Table 5: Source-traceable comparison of general restoration and editing controllers. "Preprint" denotes publication status as checked on 15 August 2026.

Work and venue/status	**Controller**	**Execution**	**Feedback**
RestoreAgent [19] NeurIPS 2024	MLLM degradation/model planner	Ordered specialist model calls	Source-defined sequence score and execution history
4KAgent [26] NeurIPS 2025	VLM–IQA perception agent	Quality-driven mixture of experts	Recursive execution–reflection
Q-Agent [64] Preprint 2025	Quality-driven CoT agent	Specialist restoration algorithms	Objective IQA-guided greedy decisions
OPERA [66] Preprint 2026	RL-optimised planner	Jointly trained restoration tools	Final restoration quality
DiTTo [65] Preprint 2026	Order-aware agent	Simulator-supported specialist tools	Degradation identification and action ordering
Restore-R1 [52] CVPR Findings 2026, P1/B	Lightweight visual actor–critic policy	Specialised restoration functions	Per-step DeQA-Score improvement
PhotoAgent [68] ICML 2026 oral	Tree-search aesthetic planner	Multi-step editing actions	Learned aesthetic reward and visual feedback
PhotoArtAgent [67] Preprint 2025	Language-model artist agents	Retouching parameters	Iterative reflection
RetouchIQ [69] CVPR 2026	RL-trained MLLM agent	Lightroom controls	Generalist retouching reward
FAPE-IR [51] CVPR 2026	Frozen MLLM planner	LoRA-MoE diffusion executor	Restoration objective across seven tasks
RAR [27] CVPR 2026, boundary	Learned latent controller	End-to-end latent iterations	Learned assessment without an external MLLM tool loop

Table 6: Source-traceable comparison of application-specific restoration controllers.

Work and venue/status	**Controller**	**Execution**	**Feedback**
JarvisIR [28] CVPR 2025	Driving-oriented MLLM controller	Restoration modules before perception	Downstream perception utility
AgentMRI [29] JIIM 2026, online 2025	Medical VLM agent	Three CycleGAN correction paths	Consensus-based corruption classification
EpiAgent [33] CVPR 2026	Observe–conceive planner	Inscription-restoration tools	Execute–reevaluate refinement
SIMBA [32] Preprint 2026	Language-to-workflow agent	JSON-schema scientific tools	Input validation and execution response
FMIRAgent [62] Research Square 2025	Self-explaining MLLM assistant	Microscopy restoration algorithms	User or simulated iterative feedback
RIR-Agent [63] ESWA 2026	MLLM perception and LLM planner	Remote-sensing toolbox	Adaptation, rollback, or manual feedback

6 Memory-Augmented and Search-Guided Evolving Agents

P4 systems retain information beyond the current action and use it to guide later restoration decisions. The retained state may be an execution trace, a reusable experience document, a search tree, or a causal memory. The following discussion separates these mechanisms because the sources implement different search and memory procedures.

6.1 Experience and Episodic Memory

AgenticIR [20] divides its workflow into perception, scheduling, execution, reflection, and rescheduling. It constructs referenceable documentation through self-exploration and can retrieve that experience for later inputs. The tool pool covers eight degradation types, with three to six specialised models per type. The paper describes rollback and rescheduling as depth-first tree search, a procedure distinct from Monte Carlo Tree Search and UCT-style rollouts.

SEAR [41] combines an Intuitive Executor with a Deliberate Planner. It uses episodic state fingerprints for memory retrieval and P-MCTS for planning. In the paper's Group B ablation, the full system averages 8.15 tool calls compared with 16.75 without memory.

6.2 Search-Guided Decision Making

Search is useful when restoration order is non-commutative and a locally attractive operation can harm later recovery. The sources implement three distinct forms.

- AgenticIR performs rollback-guided depth-first exploration and records successful experience [20].
- SEAR uses Pruning-Aware Monte Carlo Tree Search (P-MCTS) in its deliberate pathway and can accept retrieved shortcuts in its intuitive pathway [41].
- PhotoAgent applies tree search to long-horizon aesthetic editing, a different objective from degradation-removal tool routing [68].

Each algorithm is described using the search formulation reported by its source.

6.3 Causal Memory

Causal-AgentIR [42] uses multiple roles to update and reason over a causal memory. Its documented memory operations include add, update, merge, reinforce, ignore, and discard. This structure is intended to retain relations among degradations, restoration actions, and outcomes in addition to retrieving similar past trajectories. The paper reports a 35.55-dB average PSNR for its own all-in-one evaluation protocol. This number uses a different setting from the MiO100-derived group protocols used by AgenticIR and SEAR.

The evidence supports two conclusions. First, reusable experience reduces redundant planning in the SEAR ablation setting. Second, causal memory broadens the representation of prior experience beyond nearest-neighbour retrieval. Common data,

Table 7: Verified comparison of memory/search systems. Numerical values retain the source protocol.

Work	Status	Search/ control	Memory	Source-reported example
AgenticIR [20]	ICLR 2025	Depth-first rollback and rescheduling	Self-explored referenceable experience	MiO100 Group A: 21.04 dB PSNR and 0.6818 SSIM
SEAR [41]	Preprint 2026	Intuitive executor and deliberate P-MCTS	Episodic state fingerprints	MiO100 Group A: 21.8042 dB PSNR and 0.6961 SSIM. Group B memory ablation: 8.15 versus 16.75 tool calls
Causal-AgentIR [42]	Preprint 2026	Multi-agent causal reasoning	Self-evolving causal memory	All-in-one five-task average: 35.55 dB PSNR

tools, budgets, and evaluation code are still needed to measure cold-start behaviour and cross-system quality.

7 Multi-Agent and Heterogeneous Collaborative Systems

P5 distributes control across agents with distinct roles. The sources describe scheduler and specialist designs, fast and slow routing, and domain-specific coordination.

7.1 Scheduler and Specialist Agents

MAIR [38] contains a scheduler and seven expert agents. They are DeJPEG, DeBlur, DeNoise, SuperRes, DeRain, DeHaze, and LightEnh. The scheduler diagnoses the input and activates relevant experts in sequence. Within an expert, candidate restoration models can be tried, reflected on, and compared pairwise when necessary. Thus MAIR has one scheduler and seven listed experts. Candidate evaluation occurs within the expert workflow.

The paper reports 35.42 seconds and 1.82 model invocations for MAIR, compared with 63.04 seconds and 5.15 invocations for AgenticIR in its efficiency comparison. These figures are tied to the authors' environment and test protocol. Its paired real-world evaluation reports 21.67 dB PSNR.

7.2 Fast–Slow Collaboration

Hybrid Agents [39] uses three roles, FastAgent, SlowAgent, and FeedbackAgent. FastAgent is a lightweight language model (Llama-3.2-1B-Instruct) used through in-context learning. FastAgent is distinct from a fixed look-up table. FastAgent handles direct restoration instructions, whereas SlowAgent reasons about vague requests and

FeedbackAgent evaluates the result. For direct prompts, the reported fast route uses approximately 12% of SlowAgent's runtime. Table 1 of the source reports task-specific average agent inference times of 0.08–0.13 seconds for this route and 0.75–1.05 seconds when it is disabled. A reported 72.9% FastAgent success rate concerns the denoising condition.

7.3 Domain-Specific Coordination

MAPGR [34] applies multi-agent prompt guidance with residual diffusion to mural restoration. Its prompts, damage representation, and evaluation are specific to cultural-heritage restoration.

CLEAR [70] is an adjacent dual-process remote-sensing system that decides whether restoration is needed before maritime object detection. The verified paper reports 86.92% mAP50, an improvement of 7.74 percentage points over its stated baseline, and an overall F1 improvement of 9.15 percentage points. Its 11.8 FPS practical estimate assumes a 5% trigger rate. These performance and efficiency quantities come from distinct source-defined settings.

Table 8: Verified comparison of representative multi-agent or heterogeneous collaborative systems.

Work	**Venue/ year**	**Roles**	**Coordination**	**Source-reported example**
MAIR [38]	IJCV 2026	Scheduler and seven experts	Sequential scheduling and candidate comparison	35.42 s, 1.82 invocations, and paired real-world PSNR of 21.67 dB
Hybrid Agents [39]	CVPR 2026	Fast, Slow, Feedback	Request-dependent fast/slow execution and feedback	Random direct prompts: about 12% runtime, with 0.08–0.13 s versus 0.75–1.05 s when the fast route is disabled
MAPGR [34]	npj Heritage Science 2026	Visual reasoning, prompt, and execution roles	Prompt-guided dual-stream residual diffusion	Evidence propagation with low-confidence abstention
CLEAR [70]	Remote Sensing 2026	Fast/slow processing	Conditional restoration before detection	Overall mAP50 of 86.92%. The 11.8 FPS estimate assumes a 5% trigger rate

Evaluation of multi-agent systems requires reporting the number and size of controllers, expert-model executions, communication rounds, and the downstream metric. In particular, MAIR's scheduler–expert design, Hybrid Agents' request-routing design, and CLEAR's conditional preprocessing solve different control problems and require separate quality and latency interpretations.

8 Benchmarks, Evaluation Protocols, and System Metrics

Agentic image restoration requires evaluation at two levels, the restored output and the decision process that produced it. The first level uses fidelity, perceptual, or downstream-task metrics. The second uses tool calls, planning steps, latency, resource consumption, and failure recovery. A central finding of this review is that the current literature uses several mixed-degradation protocols with different test sets and tool pools. Composite-degradation results are therefore interpreted within their source protocols.

8.1 Agent Evaluation Suites and Execution Sandboxes

ImagingBench [71] covers 20 computational-imaging tasks in Expert, Planner, and Forward settings, comparing fixed or planner-guided inverse reconstruction with forward-system consistency checks. Imaging-101 [72] covers 57 expert-verified computational-imaging tasks and evaluates coding agents through planning, function-level unit tests, and end-to-end reconstruction. These suites complement output-only image benchmarks by testing whether a proposed plan is executable and physically meaningful. They are evaluation resources and therefore sit outside the P1–P5 controller taxonomy.

8.2 Datasets and Protocol Families

AIR agents are benchmarked across natural, medical, remote-sensing, and specialized domain datasets. Table 9 summarizes the principal datasets and protocol families used in the reviewed literature. Labels for mixed-degradation tests refer to study-specific protocols rather than a single standardized benchmark.

Table 9: Taxonomy of Benchmark Datasets Utilized in Agentic Image Restoration Literature.

Dataset	Modality / Domain	Degradation Types	Primary Evaluation Focus	Representative AIR Works
MiO100-derived mixed protocols [20, 27, 38, 39, 42, 52]	Natural Images	Composite (Noise + Blur + Haze + Rain + JPEG)	Multi-degradation tool routing & backtracking	RAR, Restore-R1, AgenticIR, MAIR, Hybrid Agents, Causal-AgentIR
DIV2K and high-resolution derivatives [26, 37, 73]	High-Res Natural	Downscaling (×2, ×4, ×8), Blur, Artefacts	Recursive tile-based super-resolution	RL-Restore, 4KAgent
MiO100 [20, 26, 38, 52, 65]	Natural Images	Synthetic Single / Composite Degradations (SR + Blur + Noise + JPEG + Rain + Haze + Low-light)	Degradation perception, restoration-order planning & multi-tool coordination	AgenticIR, MAIR, 4KAgent, Restore-R1, DiTTo
CDD-11 [50]	Natural Images	Composite (Low-light + Haze + Rain + Snow)	Composite degradation perception & joint restoration	OneRestore
MIT-Adobe 5K [74]	Aesthetic Photography	Poor exposure, color distortion, low contrast	Continuous slider tuning & aesthetic preference	Distort-and-Recover
Test100, Test1200, Rain100H, Rain100L [42]	Natural Images / Rainy Scenes	Rain Streaks (Varying Density & Patterns)	Single-image deraining & robustness to diverse rain patterns	Causal-AgentIR
Snow100K, SRRS, CSD [17, 42, 54]	Natural Images / Snowy Scenes	Snow Particles / Streaks (Varying Size & Density)	Single-image desnowing across diverse snow conditions	PaAgent, InstructIR, Causal-AgentIR
RESIDE, SOTS [16, 42]	Indoor & Outdoor Natural Images	Haze (Atmospheric Scattering)	Single-image dehazing across indoor/outdoor scenes	PromptIR, Causal-AgentIR
GoPro, HIDE, RealBlur [17, 42, 75, 76]	Dynamic / Human-centric Natural Scenes	Motion Blur (Camera, Object & Human Motion)	Dynamic-scene & human-aware motion deblurring	Causal-AgentIR

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Dataset	Modality / Domain	Degradation Types	Primary Evaluation Focus	Representative AIR Works
BSD68, Urban100, Kodak24, SIDD [16, 17, 42, 44, 54, 77]	Natural & Urban Images	Gaussian Noise (Multiple Noise Levels)	Image denoising across noise levels & structural complexity	PixelRL, PaAgent, PromptIR, InstructIR, Causal-AgentIR
LOL [17, 39, 42, 78]	Natural Images / Low-Light Scenes	Low Illumination (Often with Noise)	Low-light enhancement & illumination/detail restoration	Hybrid Agents , Causal-AgentIR
fastMRI and related MRI protocols [29, 47, 79–81]	Medical MRI	Study-specific 4×–6× k-space undersampling and reconstruction corruption	Sampling optimization, reconstruction, and quality-driven correction	DUAL, intelligent-agent planning, KSRO, AgentMRI
AAPM Low-Dose CT [46, 53, 82]	Medical CT	Low-photon Poisson noise, streak artefacts	Iterative reconstruction parameter tuning	Shen *et al.*, Patwari *et al.*
nuScenes and HRSC-Robust [28, 70, 83]	Autonomous Driving / Maritime Remote Sensing	Fog, Rain, Snow, low light, and other adverse conditions	Downstream driving and ship-detection performance	JarvisIR, CLEAR

8.3 Source-Traceable Quantitative Examples

Table 10 reports source-traceable examples whose values were checked against the corresponding primary papers. They illustrate what each source reports. Rows with different protocols represent separate comparisons.

Table 10: Representative quantitative results grouped by benchmark protocol.

Work	Dataset / setting	Primary metric	Result
RL-Restore [37]	DIV2K / Mild (unseen)	PSNR / SSIM	28.04 dB / 0.6498
	DIV2K / Moderate	PSNR / SSIM	26.45 dB / 0.5587
	DIV2K / Severe (unseen)	PSNR / SSIM	25.20 dB / 0.4777
PixelRL [44]	BSD68 / $\sigma = 15$	PSNR	31.49 dB
	BSD68 / $\sigma = 25$	PSNR	28.94 dB
	BSD68 / $\sigma = 50$	PSNR	25.95 dB
Path-Restore [45]	Darmstadt Noise Dataset / Path-Restore-Ext	PSNR / SSIM	39.72 dB / 0.9591
	SIDD	PSNR / SSIM	38.21 dB / 0.946
Distort-and-Recover [43]	MIT-Adobe 5K expert-C / RANDOM 250	mean $\mathcal{L}^2$ error / SSIM	10.99 / 0.905
	MIT-Adobe 5K expert-C / distort-and-recover scheme	mean $\mathcal{L}^2$ error / SSIM	12.15 / 0.910
PaAgent [54]	CSD	PSNR / SSIM	37.95 dB / 0.98
	BSD68 / $\sigma = 15$	PSNR / SSIM	34.39 dB / 0.94
	BSD68 / $\sigma = 25$	PSNR / SSIM	29.36 dB / 0.85
	BSD68 / $\sigma = 50$	PSNR / SSIM	27.03 dB / 0.79
AirNet [48]	Five-condition average: BSD68 ($\sigma = 15, 25, 50$), SOTS, Rain100L	PSNR / SSIM	31.20 dB / 0.910
PromptIR [16]	Five-condition average: BSD68 ($\sigma = 15, 25, 50$), SOTS, Rain100L	PSNR / SSIM	32.06 dB / 0.913
InstructIR [17]	BSD68, Rain100L, SOTS, GoPro, LOL	PSNR / SSIM	29.55dB / 0.907
OneRestore [50]	CDD-11	PSNR / SSIM	28.72 / 0.8821
RAR [27]	MiO100 / 2 degradations, 8 combinations	PSNR / SSIM / LPIPS	20.46 dB / 0.7144 / 0.1299
	MiO100 / 2 degradations, 4 combinations	PSNR / SSIM / LPIPS	21.04 dB / 0.7326 / 0.1269
	MiO100 / 3 degradations, 4 combinations	PSNR / SSIM / LPIPS	19.33 dB / 0.6579 / 0.1489

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Work	Dataset / setting	Primary metric	Result
Restore-R1 [52]	Setting I, 5 two-degradation combinations	PSNR / SSIM / LPIPS	19.513 dB / 0.647 / 0.365
		MANIQA / CLIP-IQA / MUSIQ / DeQA	0.349 / 0.525 / 63.003 / 3.657
	Setting II, 3 two-degradation combinations	PSNR / SSIM / LPIPS	17.958 dB / 0.636 / 0.385
		MANIQA / CLIP-IQA / MUSIQ / DeQA	0.343 / 0.530 / 63.344 / 3.599
	Setting III, 4 three-degradation combinations	PSNR / SSIM / LPIPS	17.650 dB / 0.563 / 0.475
		MANIQA / CLIP-IQA / MUSIQ / DeQA	0.295 / 0.473 / 57.756 / 3.247
	Setting IV, 3 combinations of 4 or 5 degradations	PSNR / SSIM / LPIPS	16.180 dB / 0.473 / 0.564
		MANIQA / CLIP-IQA / MUSIQ / DeQA	0.233 / 0.389 / 49.810 / 2.864
AgenticIR [20]	MiO100 / 2 degradations - 8 combinations	PSNR / SSIM / LPIPS	21.04 dB / 0.6818 / 0.3148
	MiO100 / 2 degradations - 4 combinations	PSNR / SSIM / LPIPS	20.55 dB / 0.7009 / 0.3072
	MiO100 / 3 degradations - 4 combinations	PSNR / SSIM / LPIPS	18.82 dB / 0.5474 / 0.4493
MAIR [38]	MiO100 / 2 degradations, 8 combinations	PSNR / SSIM / LPIPS	21.02 dB / 0.6715 / 0.2963
	MiO100 / 2 degradations, 4 combinations	PSNR / SSIM / LPIPS	20.92 dB / 0.7004 / 0.2788
	MiO100 / 3 degradations, 4 combinations	PSNR / SSIM / LPIPS	19.42 dB / 0.5544 / 0.4142
Hybrid Agents [39]	DF2K / Gaussian noise	PSNR / SSIM, fast route enabled vs disabled	30.25 / 0.867 vs 30.63 / 0.874
	Rain100H	PSNR / SSIM, fast route enabled vs disabled	30.04 / 0.893 vs 30.03 / 0.893
	RESIDE-6K	PSNR / SSIM, fast route enabled vs disabled	29.92 / 0.960 vs 29.92 / 0.960
	LOL	PSNR / SSIM, fast route enabled vs disabled	22.60 / 0.825 vs 22.61 / 0.828
Causal-AgentIR [42]	Five-task average: Test100, Snow100K, SOTS-Indoor, GoPro, CBSD68	PSNR / SSIM	35.55 dB / 0.964

8.4 Multi-Dimensional Evaluation Metrics

Output quality.

Full-reference fidelity uses PSNR and SSIM [84]. Perceptual distance can use LPIPS [21] or DISTS [85]. Real-world no-reference evaluation may use BRISQUE [86], MUSIQ [36], MANIQA [22], or CLIP-IQA [23]. Reports need to state the metric direction and the availability of a clean reference.

Downstream utility.

Detection mAP, segmentation mIoU, recognition accuracy, or scientific measurement error can be more relevant than visual fidelity when restoration is an upstream component. Comparison requires a fixed downstream model, evaluation split, and degraded-input baseline.

Agentic execution.

A complete agent evaluation reports the mean and distribution of model/tool invocations, planning and retry counts, end-to-end latency, peak memory, token or API cost, tool failures, and the fraction of inputs for which restoration worsens the chosen quality or task metric. For code-generating agents, compilation, runtime validity, and physical constraint violations require separate reporting.

Table 11: Minimum metric families for evaluating AIR systems.

Dimension	**Examples**	**Required context**
Fidelity	PSNR, SSIM	Dataset, split, degradation, reference availability
Perception	LPIPS, DISTS, NR-IQA	Metric direction, implementation, aggregation
Utility	mAP, mIoU, OCR/scientific error	Downstream model and degraded-input baseline
Execution	Calls, steps, retries, failures	Tool registry, stopping rule, budget
Efficiency	Latency, VRAM, tokens/cost	Hardware, model/API version, concurrency
Safety	Worsening rate, hallucination, provenance	Failure definition and audit trail

Current evidence supports protocol-aware synthesis. Ablations are interpreted within a paper, while cross-system comparisons require a shared harness with fixed inputs, tools, feedback, and budgets.

Table 12: Protocol information needed for reproducible agent evaluation.

Protocol component	**What to disclose**	**Why it matters**	**Typical failure**
Input generation	Source images, degradation order/severity, random seed	Determines problem difficulty and ordering ambiguity	Calling different mixtures the same "MixDeg" benchmark
Controller	Model/version, prompt, decoding, memory initialisation	Changes diagnosis and planning behaviour	Silent model/API updates
Tools	Exact checkpoints, parameters, permitted order	Defines the executable action space	Comparing systems with unequal tool strength
Feedback	Metric implementation, reference access, stopping rule	Determines retry and termination	Test-time access to unavailable ground truth
Budget	Calls, tokens, time, hardware, parallelism	Required for efficiency claims	Reporting tool time but omitting controller time
Statistics	Repetitions, variance, confidence intervals, failure cases	Agent decisions can be stochastic	Single-run point estimates

9 Open Challenges and Strategic Future Roadmaps

Agentic image restoration remains constrained by distribution shift, acquisition physics, reasoning cost, and the risk that a generative prior introduces unsupported detail. These limitations define four closely related research directions.

Many tool-using systems expose restoration models as external functions. OPERA addresses the coordination gap between a planner and independently trained executors [66]. A physics-aware controller could additionally test candidate outputs against a known image-formation model. Forward-model residuals and uncertainty estimates would make physical inconsistency measurable. Clinical or scientific validity still requires domain-specific evaluation.

Acquisition and restoration can also be treated as a joint control problem. JarvisIR links restoration to downstream driving perception [28], while MRI policies select measurements under reconstruction feedback [47]. Future systems could choose exposure, gain, sampling, or re-acquisition actions under a fixed dose, time, energy, or motion budget. Evaluation would require a matched passive acquisition baseline under the same budget.

Memory-based systems offer a basis for controlled stress testing. Causal-AgentIR updates a structured memory through add, update, merge, reinforce, ignore, and discard operations [42]. Adversarial or curriculum-generated degradations could extend this process by searching for controller failures. The resulting synthetic corruptions would need validation against real sensor statistics and evaluation on held-out real degradations.

Auditable restoration requires a record of the source asset, controller, tool versions, parameters, ordered actions, intermediate outputs, and uncertainty. C2PA Technical Specification 2.4 provides a current mechanism for signed content credentials and provenance assertions [87]. Such provenance establishes lineage and reproducibility, while domain experts remain responsible for judging whether the restored content is scientifically or clinically valid.

10 Conclusion

This structured review organises agentic image restoration around four functional components and five controller families. The components are perception, decision, action, and reflection or memory. The controller families cover RL policies, prompt-conditioned restoration, single-MLLM tool use, memory and search, and multi-agent collaboration.

The literature documents a progression from fixed action policies to semantic planning, reusable experience, and distributed roles. The boundaries between these systems remain important. RAR is a learned latent restore–assess–repeat model [27]. AgenticIR uses depth-first rollback with referenceable experience [20]. SEAR combines an intuitive executor with deliberate P-MCTS and episodic memory [41]. These mechanisms represent distinct forms of adaptation and require separate evaluation.

Current quantitative evidence is fragmented across datasets, degradation mixtures, tool pools, metrics, and computational budgets. Progress therefore depends on fixed test-generation procedures, disclosed controller and tool versions, matched action spaces, complete execution budgets, repeated trials, and explicit failure reporting. Under these conditions, agentic control can be evaluated for adaptive diagnosis, traceable decisions, recovery from poor intermediate results, and alignment with user or downstream utility.

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